CONDENSED MATHEMATICS MEETS TOPOLOGICAL QUANTUM FIELD THEORY: A FORMALLY VERIFIED FOUNDATION

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ABSTRACT. We present a formally verified investigation into using condensed abelian groups (Clausen–Scholze) as a target category for topological quantum field theories, replacing topological vector spaces which fail to be abelian. Through successive rounds of AI-assisted formal verification using Lean 4, Mathlib v4.28.0 [The24], Harmonic's Aristotle system [ABD+25], Claude, and OpenAI Codex, we establish: (1) a symmetric monoidal structure on CondensedAb via sheaf localization; (2) an abstract TQFT transfer theorem; (3) an embedding functor SemiNormedGrp $\to$ CondensedAb; (4) a complete structural profile of this embedding—faithful and left exact, but not full, not conservative, and not right exact—with explicit counterexamples; (5) a hierarchy of algebraic cobordism-presentation quotients culminating in a symmetric monoidal category and a strong braided monoidal interpretation for every commutative Frobenius datum in a symmetric target; (6) a rank-n diagonal Frobenius theory whose torus and defined connected genus words act by $n\,\mathrm{id}$, both in the base presentation and through the underlying functor of the packaged strong monoidal, braided theory on the symmetric algebraic quotient; and (7) an abstract ribbon-category layer in which quantum trace is cyclic, the S-pairing is symmetric, and quantum dimension is multiplicative.

The project comprises approximately 4,700 lines of Lean 4 across 12 source files, with **no remaining sorry placeholders** and 0 custom axioms: every formal theorem reported here is machine-checked. The cobordism constructions are algebraic presentations; no equivalence with the geometric oriented bordism category, universal classification theorem, or non-compact Chern–Simons construction is claimed.

Contents

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1. Introduction

1.1. The Homological Obstruction in Infinite-Dimensional TQFT. A topological quantum field theory, in Atiyah's formulation [Ati88], is a symmetric monoidal functor $Z : \mathrm{Cob}_{d+1} \to C$ from a cobordism category to some symmetric monoidal target category C. For finite-dimensional state spaces (Dijkgraaf–Witten [DW90], Reshetikhin–Turaev [RT91], and Turaev–Viro theories), the target $C = \mathrm{Vect}$ works perfectly.

The physically interesting theories force infinite-dimensional state spaces. Chern–Simons theory [Wit88] with non-compact gauge group $\mathrm{SL}(2, \mathbb{C})$, quantum gravity partition functions, and path integrals over infinite-dimensional moduli all produce state spaces $Z(\Sigma)$ that cannot live in finite-dimensional Vect. The natural target becomes topological vector spaces.

Here a precise obstruction arises for homological and derived enhancements. The category TopAb of topological abelian groups is *not abelian*: a continuous bijective homomorphism need not have a continuous inverse, so a morphism can be simultaneously monic and epic without being an isomorphism. Consequently kernels, cokernels, exact sequences, and derived functors do not have the formal behavior available in an abelian category. This obstructs exact-sequence methods one may wish to combine with cutting and gluing. It is not an obstruction to the bare Atiyah–Segal axiom: an ordinary TQFT requires functorial composition and a symmetric monoidal product, not an abelian target.

Standard completed tensor products on topological vector spaces need not be exact: tensoring a short exact sequence with a topological vector space can destroy exactness. Disjoint union of boundaries still asks only for a monoidal product; exactness is an additional property needed when gluing calculations are organized through exact sequences or derived constructions. Thus topological vector spaces can serve as targets for ordinary symmetric-monoidal TQFTs while remaining poorly suited to these homological enhancements.

1.2. The Condensed Fix. Clausen and Scholze's condensed mathematics [Sch19] provides an abelian replacement for this homological setting. The category of condensed abelian groups (sheaves of abelian groups on the coherent site of compact Hausdorff spaces) is an abelian category with all limits and colimits. Every morphism has a kernel and cokernel. Every monic epic is an isomorphism.

Liquid vector spaces, a refinement of condensed modules, admit an exact completed tensor product. This was the content of the Liquid Tensor Experiment [Sch22, CT$^+$22], Lean-verified by

Commelin et al. in 2022. This exactness is valuable for homological gluing constructions, although it is stronger than the axioms of an ordinary TQFT and is not established by the present codebase.

1.3. **This Paper.** We report on a formal verification effort investigating whether CondensedAb can serve as a TQFT target category. The work proceeded through successive stages using Harmonic's Aristotle system [ABD+25], Claude (Anthropic), and OpenAI Codex for Lean 4 proof search, mathematical synthesis, API reconnaissance, and independent proof auditing.

The project produced both expected confirmations and genuine mathematical surprises. The monoidal structure on CondensedAb turned out to already exist in Mathlib. The embedding from seminormed groups turned out not to be full: a counterexample found while attempting to formalize fullness shows the obstruction, and we machine-checked it, first at the presheaf level and then for the sheaf-level functor itself. The later stages then ran an explicit Frobenius theory through the combinatorial semantics, closed the algebraic monoidal and symmetry gap in the source presentation, transported the concrete closed-word computations through the packaged symmetric-quotient theory, and completed a ribbon-category proof of quantum-dimension multiplicativity.

2. The Formalization

2.1. **Session 1: Abstract TQFT Framework.** The first session established the categorical scaffolding in the file `LiquidTQFT.lean`.

Definition 2.1. An *abstract TQFT* with source category S and target category C (both braided monoidal) is a braided monoidal functor $Z : S \to C$.

```
structure AbstractTQFT
    (S : Type*) [Category S] [MonoidalCategory S] [BraidedCategory S]
    (C : Type*) [Category C] [MonoidalCategory C] [BraidedCategory C]
      where
  Z : Functor S C
  monoidal : Z.Monoidal
  braided : Z.Braided
```

We package braided monoidal functors, a level of generality that includes the symmetric case and interfaces with non-symmetric braided categories used as algebraic input to three-dimensional constructions. Modular tensor categories underlying Reshetikhin–Turaev invariants [RT91] are a principal example of such input; the resulting oriented TQFT itself has the usual symmetric-monoidal bordism source. Lurie's cobordism hypothesis [Lur09] provides the fully extended framework in which these structures are most naturally understood.

Theorem 2.2 (Transfer, sorry-free). *If T is a TQFT valued in C and $F : C \to D$ is a braided monoidal functor, then $T \circ F$ is a TQFT valued in D.*

```
def AbstractTQFT.transfer
    (T : AbstractTQFT S C)
    (F : Functor C D) [hF : F.Monoidal] [hFb : F.Braided] :
    AbstractTQFT S D where
  Z := F.comp T.Z
  monoidal := by letI := T.monoidal; infer_instance
  braided := by letI := T.monoidal; letI := T.braided; infer_instance
```

The proof is Lean's typeclass inference composing the monoidal and braided structures. Additional sorry-free results from this session: CondensedAb is abelian, has kernels, cokernels, and finite (co)limits, and satisfies the balanced category property (mono + epi = iso).

The session also recorded structural observations for homological enhancements: in an abelian category, short exact sequences support decomposition methods one may combine with cutting and gluing. These statements take exactness as a hypothesis. The substantive analytic claim that the liquid tensor product is exact depends on the LTE and is not verified in this codebase; it is an additional strengthening, not part of the ordinary TQFT axioms.

2.2. **Session 2: Manual Monoidal Construction.** The second session (`CondensedMonoidal.lean`) attempted a direct construction of the monoidal structure on CondensedAb, defining the tensor product as sheafification of the pointwise tensor:

$$F \otimes_{\text{cond}} G = \text{sheafify}(F \otimes_{\text{presheaf}} G).$$

This yielded a complete `MonoidalCategoryStruct` (sorry-free) and 5/10 `MonoidalCategory` axioms. The remaining 5 reduced to two "sheafification comparison" isomorphisms:

$$\text{sheafify}(P \otimes Q) \cong \text{sheafify}(\text{sheafify}(P)^{\text{val}} \otimes Q).$$

This analysis was essential for the breakthrough in Session 3. As this manual construction was entirely superseded, it is not included in the final repository; we describe it here because the failed attempt is what motivated the search that found the localization approach.

2.3. **Session 3: The Localization Breakthrough.** The third session discovered that `Mathlib.CategoryTheory.Sites.Monoidal` [Theb] already contained the complete infrastructure.

Theorem 2.3 (Sorry-free)**.** CondensedAb *is a symmetric monoidal category.*

The construction assembles four existing Mathlib components:

1. **Presheaf monoidal structure**: pointwise tensor on $\text{CompHaus}^{\text{op}} \to \text{Ab}$.
2. **Sheafification as localization**: `presheafToSheaf` is a localization functor for the class $J.W$ of local isomorphisms.
3. W **is monoidal**: $J.W$ is closed under tensoring, proved via the internal hom (monoidal closed structure), not stalks.
4. **Localized monoidal category**: general construction from [Thea].

```
instance condensedAb_W_isMonoidal :
    ((coherentTopology CompHaus).W
      (A := ModuleCat (ULift Int))).IsMonoidal :=
  GrothendieckTopology.W.monoidal

instance condensedAb_monoidalCategory :
    MonoidalCategory CondensedAb :=
  Sheaf.monoidalCategory _ _
```

Six lines, zero sorries. The internal hom argument (rather than stalks) powers the `W.IsMonoidal` proof, meaning the result holds for any closed braided target category. A direct stalk-based approach has since been formalized by Riou (Mathlib PR #35584), providing an alternative proof path.

2.4. **Session 4: The Banach Embedding.** The fourth session (`BanachEmbedding.lean`) audited Mathlib for connections between normed spaces and condensed mathematics. No packaged bridge was found in the pinned Mathlib v4.28.0 source.

Definition 2.4. For a seminormed abelian group V, the presheaf banachPresheaf V sends $S \in$ CompHaus to $C(S, V)$, the abelian group of continuous maps, with precomposition as functorial action.

Theorem 2.5 (Sorry-free)**.** banachPresheaf V *is a sheaf for the coherent topology on* CompHaus.

The proof transfers the sheaf condition from the Type-valued `yonedaPresheaf` through the forgetful functor. The main technical obstacle was universe arithmetic, bridged using `hasLimitsOfSizeShrink` and `preservesLimitsOfSize_of_univLE`.

Theorem 2.6 (Sorry-free)**.** *There exists a functor* SemiNormedGrp $\to$ CondensedAb.

2.5. **Session 5: Faithful but Not Full.** The fifth session established the central structural property of the embedding.

Theorem 2.7 (Sorry-free)**.** *The embedding* semiNormedGrpToCondensedAb : SemiNormedGrp $\to$ CondensedAb *is faithful but* ***not*** *full. Both statements are machine-checked for this single sheaf-level functor.*

Proof. Faithfulness is the injectivity of postcomposition on bounded maps, verified by evaluation at the one-point space: constant functions separate the points of any seminormed group, so distinct bounded maps induce distinct condensed morphisms.

For non-fullness, let $V = \mathbb{N} \to_0 \mathbb{Z}$ (finitely supported integer sequences) with the supremum norm. Because every nonzero integer entry has absolute value at least 1, the norm satisfies $\|a\| \geq 1$ for $a \neq 0$, so V carries the *discrete* topology; consequently every additive map out of V is continuous. The summation map $f(a) = \sum_n a_n$ is therefore a continuous additive homomorphism, but it is not bounded: the indicators w_N of $\{0, \ldots, N\}$ satisfy $\|w_N\| = 1$ while $f(w_N) = N+1$. Postcomposition by f defines a genuine morphism banachCondensed $V \to$ banachCondensed $\mathbb{Z}$ in CondensedAb (no sheafification argument is required, since both objects are already sheaves by construction). If the embedding were full, this morphism would come from a bounded map agreeing with f; evaluation at the one-point space forces pointwise agreement, contradicting unboundedness. $\square$

The formalization proceeds in two layers. `FullnessCounterexample.lean` verifies the analytic core and a self-contained presheaf-level functor E : SemiNormedGrp $\to$ PSh(CompHaus, Ab) that is faithful and not full. `SheafFullnessCounterexample.lean` then transports the counterexample (after a universe lift, under which the discreteness and norm computations are unchanged) to the sheaf-level functor itself, yielding $\neg$ semiNormedGrpToCondensedAb.Full directly. The key observation is that the sheaf condition is never needed: the source and target of the offending morphism are already sheaves, so the morphism can be built between them outright.

The discreteness of V is the engine: the condensed setting sees *all* additive maps as morphisms, while SemiNormedGrp admits only bounded ones, so the embedding cannot be full. The entire faithful-but-not-full development is sorry-free and uses no custom axioms.

Root cause. Continuous additive maps between seminormed abelian groups need not be bounded. The standard proof that "continuous linear implies bounded" requires scalar multiplication by a dense subfield. This suggests restricting the source to normed spaces over a nontrivially normed field; fullness of that variant is not formalized here.

The session also constructed SemiNormedGrp.hasFiniteProducts (sorry-free and not available in the pinned Mathlib version) and proved that the embedding preserves both finite products and equalizers, hence all finite limits, routing through the fact that the sheaf inclusion creates limits. The embedding is therefore left exact, machine-checked end to end.

2.6. **Session 6: Combinatorial Cobordisms.** The sixth session (`Cob2.lean`) formalized the algebraic structure classifying 2-dimensional TQFTs [Koc03].

Definition 2.8 (Sorry-free)**.** A *Frobenius object* in a monoidal category extends Mathlib's monoid and comonoid objects with the Frobenius relation:

$$(\mu \otimes \mathrm{id}) \circ (\mathrm{id} \otimes \delta) = \delta \circ \mu.$$

A *commutative Frobenius object* additionally requires $\mu \circ \beta = \mu$.

Theorem 2.9 (Sorry-free)**.** *In any braided monoidal category, the unit object carries a canonical commutative Frobenius algebra structure.*

We define a combinatorial category Cob_2 on the natural numbers, with morphisms generated by μ (pants), η (cap), δ (copants), ε (cup), plus composition, tensor, and swap, quotiented by the category axioms and the commutative Frobenius equations (with congruence closure). We emphasize the scope of this construction: tensor interchange, monoidal coherence on morphisms, and braiding naturality are *not* imposed, so this quotient is a preliminary presentation, coarser than a full presentation of the 2-dimensional cobordism category, and it carries no monoidal structure of its own.

Theorem 2.10 (Sorry-free)**.** *Every commutative Frobenius datum A in a braided monoidal category C induces a functor* $\mathrm{Cob}_2 \to C$ *sending n to $X^{\otimes n}$ and the generators to the corresponding structure maps of A.*

This is a first algebraic rung toward one direction of the Frobenius classification of 2d TQFTs [Koc03], formalized via a recursive interpretation of generator words together with a proof that the interpretation respects every imposed relation. The base category `Cob2Cat` intentionally remains an ordinary preliminary quotient. Session 9 below strengthens its relation to obtain lawful monoidal and symmetric quotients. Even after that strengthening, no normal-form theorem, universal classification property, or identification with geometric oriented bordisms is asserted.

2.7. **Session 7: The Full Profile of the Embedding.** Session 7 (`EmbeddingProfile.lean`) added two machine-checked negative results and completed the embedding's structural profile.

Theorem 2.11 (Sorry-free)**.** *The embedding does not reflect isomorphisms. Explicitly, the ℓ^1 and supremum norms on finitely supported integer sequences both induce the discrete topology, so the bounded identity between them realizes to an isomorphism in* CondensedAb *(the set-theoretic inverse is continuous and additive), yet it is not an isomorphism in* SemiNormedGrp*: any bounded inverse is refuted on the indicator vectors w_N of ℓ^1 norm $N+1$ and sup norm 1.*

Theorem 2.12 (Sorry-free)**.** *The embedding does not preserve epimorphisms, and in particular is not right exact. Explicitly, equip $\mathbb{R}$ with the discrete group norm $\|x\|_d = |x| + [x \neq 0]$; the bounded surjective identity onto euclidean $\mathbb{R}$ is an epimorphism in* SemiNormedGrp*, but its realization fails the local-surjectivity criterion for epimorphisms of condensed modules: the section $[0,1] \hookrightarrow \mathbb{R}$ admits no local lift, since a continuous map from a compact space to a discrete space has finite image.*

Together with the earlier results, this yields a complete profile: the embedding is **additive, faithful, and left exact, but not full, not conservative, and not right exact**. The asymmetry (left exact but not right exact) is precisely the sense in which the condensed category re-topologizes the normed world: quotients that are invisible to bounded maps acquire genuine local structure on the condensed side.

2.8. **Session 8: A Concrete Diagonal Frobenius Theory.** The file `DijkgraafWitten.lean` supplies the first nontrivial input to the combinatorial semantics. For each n, let

$$A_n = \mathbb{Z}^{\mathrm{Fin}(n)}$$

with pointwise multiplication. On the standard basis e_i, the structure maps are

$$\eta(1) = \sum_i e_i, \qquad \delta(e_i) = e_i \otimes e_i, \qquad \varepsilon(x) = \sum_i x_i.$$

The construction uses `Pi.basisFun`, `Basis.constr`, and Mathlib's tensor product API. Associativity, the unit and counit laws, coassociativity, both Frobenius equations, and commutativity are proved by basis reduction.

Theorem 2.13 (Diagonal torus and genus evaluation, sorry-free)**.** *For the rank-n datum* `frobZn`*, the closed word*

$$\varepsilon \circ \mu \circ \delta \circ \eta$$

evaluates on $1 \in \mathbb{Z}$ to n. Equivalently, its image is multiplication by n on the monoidal unit. More generally, every connected closed genus-g presentation word defined in the file has the same value n.

The generic calculation expresses a genus word as unit, a power of the handle operator $\mu \circ \delta$, and counit. For A_n the handle operator is the identity, so all handle powers collapse and the remaining coordinate sum is n. This is a concrete finite-state Frobenius toy theory. It is not the conventional finite-group Dijkgraaf–Witten state-sum construction [DW90], and until the geometric comparison is established its theorem is an evaluation of algebraic presentation words rather than a diffeomorphism invariant of smooth surfaces. Session 10 transports these same scalar morphism equalities through the symmetric algebraic quotient without changing this geometric limitation.

2.9. **Session 9: Lawful Monoidal and Symmetric Cobordism Quotients.** The files `Cob2Monoidal.lean` and `Cob2Symmetric.lean` close the algebraic coherence gap identified in Session 6. The first enlarges the relation by tensor identities, interchange, and associator and unitor naturality. The resulting quotient is a lawful monoidal category; its pentagon and triangle equations follow from the arithmetic equality transports used for structural morphisms.

Theorem 2.14 (Strong monoidal interpretation, sorry-free)**.** *Every commutative Frobenius datum in a braided monoidal category induces a strong monoidal functor from the strengthened monoidal quotient, and precomposition with the canonical quotient functor recovers the original interpretation.*

The symmetric rung further imposes swap naturality, swap involutivity, and both braided hexagons.

Theorem 2.15 (Symmetric algebraic source, sorry-free)**.** *The strengthened quotient is a symmetric monoidal category. Every commutative Frobenius datum in a symmetric monoidal target induces a strong braided monoidal functor from this quotient. Passing through the symmetric quotient recovers the monoidal interpretation.*

Lean packages this result as `toSymmetricTQFT2d`. Session 10 instantiates this package with the diagonal datum and computes its transported closed presentation classes.

This completes the intended algebraic interchange-and-braiding rung. It does *not* prove that the quotient is equivalent to the category of compact oriented smooth surfaces up to boundary-preserving diffeomorphism, nor does it establish the universal free-commutative-Frobenius-object theorem. Those are separate geometric and classification statements.

2.10. **Session 10: The Diagonal Theory on the Symmetric Quotient.** The new transport module composes the two existing ordinary quotient functors,

$$\begin{aligned} \mathrm{Cob2Cat} &\longrightarrow \mathrm{Cob2MonoidalObj} \\ &\longrightarrow \mathrm{Cob2SymmetricObj}, \end{aligned}$$

to define `Cob2.toSymmetricQuotient`. For every commutative Frobenius datum A in a symmetric target, interpretation after this composite is proved equal, as an ordinary functor, to `A.toCob2Functor`; on raw representatives the two maps are definitionally equal. No monoidal structure on this composite quotient functor is asserted.

For the diagonal datum, `frobZnSymmetricTQFT` packages `toSymmetricTQFT2d`. Its underlying functor is strong monoidal and braided, with symmetric source and target. The presentation classes `symmetricTorus` and `symmetricGenus g` are defined by transporting the already verified base classes, and `symmetricGenus 1 = symmetricTorus`.

Theorem 2.16 (Symmetric-quotient diagonal evaluation, sorry-free). *For every $n, g \in \mathbb{N}$, if Z_n denotes the underlying functor of `frobZnSymmetricTQFT n`, then*

$$Z_n(\texttt{symmetricTorus}) = (n : \mathbb{Z})\,\mathrm{id}_{\mathbf{1}}, \qquad Z_n(\texttt{symmetricGenus } g) = (n : \mathbb{Z})\,\mathrm{id}_{\mathbf{1}}.$$

In particular, both underlying module morphisms evaluate at $1 \in \mathbb{Z}$ to n.

These proofs transport the scalar morphism equalities from `DijkgraafWitten.lean`; they do not repeat the basis or tensor-product calculation. The result is a concrete strong monoidal, braided algebraic theory between symmetric categories. It does not identify the source with geometric oriented bordisms or establish classification, diffeomorphism, or gluing invariance. The `TQFT2d` package stores `Functor.Monoidal` and `Functor.Braided`; Lean has no additional field named `Functor.Symmetric` here.

2.11. **Session 11: Ribbon Categories and Quantum Dimension.** The root-level file `Ribbon.lean` develops an input-side abstraction relevant to three-dimensional TQFT. A balanced monoidal category is equipped with a natural twist satisfying the balancing equation, and a ribbon category additionally has chosen right duals with twist compatible with dualization. Every symmetric right-rigid monoidal category receives the identity-twist ribbon structure.

The file defines the quantum trace $\mathrm{qTr}(f)$, quantum dimension $\mathrm{qdim}(X) = \mathrm{qTr}(\mathrm{id}_X)$, and the S-pairing as the quantum trace of the double braiding. It proves trace cyclicity by sliding morphisms through the cup–braid–twist–cap diagram.

Theorem 2.17 (Quantum-dimension multiplicativity, sorry-free). *In every ribbon category,*

$$\mathrm{qdim}(X \otimes Y) = \mathrm{qdim}(X) \circ \mathrm{qdim}(Y) \quad \textit{in } \mathrm{End}(\mathbf{1}).$$

Moreover the S-pairing is symmetric.

The multiplicativity proof constructs the reversed tensor-product exact pairing, verifies both triangle identities, factors the balanced evaluation using three Yang–Baxter rewrites, and moves the resulting scalar through the tensor factors. This is useful ribbon input infrastructure, but it does not construct a modular tensor category, prove nondegeneracy of the S-pairing, or define Reshetikhin–Turaev 3-manifold invariants.

3. Results Summary

3.1. **Theorem Inventory.**

3.2. **Completeness of the Formalization.** The formalization contains no executable `sorry` or `admit` placeholders and introduces no custom axioms: every formal theorem reported here is fully machine-checked. Dependency audits for the new headline theorems report only Lean's standard `propext`, `Classical.choice`, and `Quot.sound`.

Both limit-preservation results for the embedding were proved through a common architecture. `PreservesFiniteProducts` routes through the fact that the sheaf inclusion creates limits. `PreservesEqualizers` follows the same path: the canonical equalizer fork in SemiNormedGrp (vertex $\ker(f - g)$ with the subspace topology, already available in Mathlib's kernels file) is sent to a limit fork of types by each $C(S, -)$, since an equalizing continuous map lands in the kernel and corestricts continuously; the statement then lifts pointwise through evaluation and through the sheaf inclusion. Consequently the embedding preserves all finite limits: **the embedding** SemiNormedGrp $\to$ CondensedAb **is left exact, machine-checked end to end.**

The base functor toCob2Functor was completed by recursively interpreting raw generator words as morphisms between tensor powers. The tensor case uses comparison isomorphisms $X^{\otimes(a+c)} \cong X^{\otimes a} \otimes X^{\otimes c}$, built by induction from associators and the right unitor; the monoid and comonoid

Result	**File**	**Status**
CondensedAb is abelian	LiquidTQFT.lean	Sorry-free
Mono + epi = iso	LiquidTQFT.lean	Sorry-free
MonoidalCategory CondensedAb	LiquidTQFT.lean	Sorry-free
BraidedCategory CondensedAb	LiquidTQFT.lean	Sorry-free
SymmetricCategory CondensedAb	LiquidTQFT.lean	Sorry-free
`AbstractTQFT.transfer`	LiquidTQFT.lean	Sorry-free
W.IsMonoidal for coherent topology	MonoidalViaLoc.lean	Sorry-free
`presheafToSheaf` is monoidal	MonoidalViaLoc.lean	Sorry-free
`banachPresheaf` V is a sheaf	BanachEmbedding.lean	Sorry-free
SemiNormedGrp $\to$ CondensedAb functor	BanachEmbedding.lean	Sorry-free
Embedding faithful (sheaf level)	BanachEmbedding.lean	Sorry-free
Embedding not full (sheaf level)	SheafFullnessCounterexample.lean	Sorry-free
Presheaf functor E faithful, not full	FullnessCounterexample.lean	Sorry-free
Embedding preserves finite products	BanachEmbedding.lean	Sorry-free
Embedding preserves equalizers	BanachEmbedding.lean	Sorry-free
Embedding is left exact	BanachEmbedding.lean	Sorry-free
Embedding does not reflect isos	EmbeddingProfile.lean	Sorry-free
Embedding does not preserve epis	EmbeddingProfile.lean	Sorry-free
SemiNormedGrp.hasFiniteProducts	BanachEmbedding.lean	Sorry-free
`FrobeniusObj`, `CommFrobeniusObj`	Cob2.lean	Sorry-free
Trivial Frobenius on $\mathbf{1}_C$	Cob2.lean	Sorry-free
Frobenius datum $\Rightarrow$ functor from Cob_2	Cob2.lean	Sorry-free
Cob_2 category	Cob2.lean	Sorry-free
Rank-n diagonal Frobenius datum	DijkgraafWitten.lean	Sorry-free
Torus and genus-word evaluations equal n	DijkgraafWitten.lean	Sorry-free
Lawful monoidal presentation quotient	Cob2Monoidal.lean	Sorry-free
Strong monoidal Frobenius interpretation	Cob2Monoidal.lean	Sorry-free
Symmetric monoidal presentation quotient	Cob2Symmetric.lean	Sorry-free
Strong braided monoidal interpretation	Cob2Symmetric.lean	Sorry-free
Base-to-symmetric interpretation compatibility	DijkgraafWittenSymmetric.lean	Sorry-free
Packaged symmetric torus/genus maps equal n id	DijkgraafWittenSymmetric.lean	Sorry-free
Quantum-trace cyclicity and S-symmetry	Ribbon.lean	Sorry-free
Quantum-dimension multiplicativity	Ribbon.lean	Sorry-free

TABLE 1. Theorem inventory for the machine-checked development. The faithful-but-not-full result (Theorem 2.7) is verified for the sheaf-level embedding itself. The cobordism quotients are algebraic presentations; geometric identification remains future work.

generators are conjugated by unitors to match this convention. Soundness over the original presentation equations is proved constructor by constructor, after which the functor descends via `Quotient.lift`.

The two strengthened quotients then close the algebraic coherence gap. `Cob2MonoidalRel` adds tensor identity, interchange, and structural naturality; coherence of the same `powAdd` comparisons supplies a strong monoidal interpretation. `Cob2SymmetricRel` adds swap naturality, involutivity, and both hexagons; interpretation into a symmetric target is sound and braided. The compatibility theorems show that passing from the base quotient through the monoidal and symmetric rungs recovers the earlier semantics. The composite `Cob2.toSymmetricQuotient` now packages both steps, and interpretation after it is proved equal to the base semantics for every commutative Frobenius datum. What remains open is geometric and universal: no theorem identifies this quotient with oriented smooth bordisms or proves the full Frobenius classification equivalence.

The diagonal basis calculation is performed once in `DijkgraafWitten.lean`. `DijkgraafWittenSymmetric.lean` reuses it to prove the scalar morphism equalities through the underlying functor of the packaged symmetric-quotient theory; no basis-level computation is duplicated. The ribbon proof builds and verifies the reversed tensor pairing explicitly; a six-crossing braid equality, resolved by three Yang–Baxter moves, is the key step in factoring the quantum cap and proving multiplicativity.

3.3. **Key Findings.** **Finding 1.** The monoidal structure on CondensedAb was already in Mathlib. The iterative process (axiom $\to$ manual construction $\to$ localization discovery) demonstrates the difficulty of navigating large formal libraries.

Finding 2. No packaged bridge was found between Mathlib's `Analysis.NormedSpace` and `Condensed` components in v4.28.0. The project supplies a concrete connection.

Finding 3. Fullness fails for the embedding (Theorem 2.7), machine-checked for the sheaf-level functor itself. The condensed setting sees strictly more morphisms than the normed category, because the sup-norm topology on integer sequences is discrete and admits unbounded continuous additive maps.

Finding 4. SemiNormedGrp.hasFiniteProducts was not available in the pinned Mathlib v4.28.0 source.

Finding 5. The Frobenius-generator semantics extends to a lawful symmetric monoidal algebraic source and a strong braided monoidal interpretation. The canonical composite from the base quotient to the symmetric quotient is semantically compatible with the original interpretation for every commutative Frobenius datum. This verifies a substantial algebraic direction of the 2d TQFT/Frobenius correspondence [Koc03], while leaving its universal and geometric identifications open.

Finding 6. The rank-n diagonal Frobenius theory is executable mathematics: the torus and every connected genus word in the defined family act by multiplication by n, and these scalar morphism equalities persist through the underlying functor of the packaged strong monoidal, braided theory on the symmetric algebraic quotient.

Finding 7. Ribbon-category string-diagram identities can be proved directly against Mathlib's associator, braiding, and exact-pairing APIs. In particular, quantum dimension is multiplicative and the S-pairing is symmetric without assuming modularity.

4. Roadmap for Further Development

4.1. **Near-term.** Compute disconnected tensor products of the transported closed presentation classes; prove useful normal forms for connected presentation words; and instantiate the finite diagonal module inside CondensedAb through the discrete embedding.

4.2. **Medium-term.** Prove the universal property of the symmetric algebraic quotient and compare it with a combinatorial model of oriented 2-dimensional bordisms. On the analytic side, define an appropriate category of normed spaces over a valued field and determine which fullness or exactness statements survive after imposing the hypotheses of the open mapping theorem.

4.3. **Long-term.** Formalize compact smooth manifolds with boundary, gluing, and diffeomorphism classes sufficiently to identify the algebraic source with geometric 2Cob. Develop modular tensor category input and surgery/Kirby-move infrastructure for Reshetikhin–Turaev theory; `Ribbon.lean` supplies only the first input-side layer. Formalize the projective tensor product of normed spaces and connect to liquid tensor exactness.

4.4. **Frontier.** Construct mathematically rigorous non-compact Chern–Simons state spaces and functorial gluing before attempting their formalization in liquid or nuclear targets. A parallel frontier is Costello–Gwilliam [CG17] factorization algebras in the condensed setting.

5. Methodology

The work proceeded through successive rounds of AI-assisted formal verification. A typical round consisted of: (1) scoping a concrete mathematical target; (2) reconnaissance of the exact Mathlib API and categorical conventions; (3) construction and proof search; and (4) independent compilation, placeholder scanning, and axiom-dependency auditing. Aristotle and Claude supported earlier rounds; OpenAI Codex completed and independently audited the ribbon multiplicativity proof, the symmetric-quotient transport, and repository integration.

The prompts evolved from exploratory to targeted. The most productive prompts front-loaded reconnaissance (searching Mathlib for existing infrastructure) before attempting construction. The Session 3 breakthrough directly resulted from this "search before build" approach.

5.1. Tools.

- **Lean 4** with **Mathlib v4.28.0** [The24] for formal verification.
- **Harmonic Aristotle** [ABD+25] for proof search and Mathlib API discovery.
- **Claude (Anthropic)** for mathematical synthesis, prompt engineering, and documentation.
- **OpenAI Codex** for proof completion, independent auditing, and repository integration.
- All formal results independently verifiable via `lake build`; headline proofs were additionally checked with `#print axioms`.

6. Conclusion

Successive rounds of AI-assisted formal verification, starting from a blank slate, produced approximately 4,700 lines of Lean 4 across 12 source files with no remaining sorry placeholders and 0 custom axioms. The central results are machine-checked end to end: CondensedAb is a symmetric monoidal abelian category; the TQFT transfer theorem holds; the seminormed-group realization is faithful and left exact but not full, not conservative, and not right exact; the Frobenius presentation admits lawful monoidal and symmetric algebraic quotients with strong monoidal semantics; the rank-n diagonal theory sends its transported symmetric torus and defined connected genus classes to n times the identity under the underlying functor of the packaged strong monoidal, braided algebraic theory; and quantum dimension is multiplicative in every ribbon category.

A notable result was machine-checking that the embedding SemiNormedGrp $\to$ CondensedAb is faithful but not full, with both directions verified for the sheaf-level functor itself. This is a precise mathematical boundary: the condensed setting sees more morphisms between normed groups than the normed category itself does, because the sup-norm topology on integer sequences is discrete and admits unbounded continuous additive maps. Normed vector spaces over a valued field provide the natural next setting because continuity of linear maps then controls boundedness; that variant is not formalized here.

The remaining barriers are now sharply separated by scale. The next algebraic step is a universal property and normal forms for the symmetric quotient. The next geometric step is a formal category of smooth surfaces with boundary and a comparison with that quotient. Three-dimensional Reshetikhin–Turaev theory additionally requires modular tensor categories, surgery presentations, and Kirby moves. The motivating non-compact Chern–Simons target lies beyond these: its infinite-dimensional state spaces and gluing theory must first be constructed rigorously on paper before the liquid or nuclear machinery can be applied.

Code Availability. The paper and a synchronized Lean 4 snapshot are available at `https://github.com/seanm27lol/Liquid-TQFT-lean-CMCM`. The canonical implementation repository is `https://github.com/seanm27lol/Liquid-TQFT-CMCM-cont.`; the results reported here correspond to commit `b350215`. Both build against Mathlib v4.28.0.

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